Point and probabilistic forecasts of wind with complex ARIMA: design and results of the second experiment

Wind Forecasting with Complex ARIMA — working notes

26 October 2026

1 Purpose

The paper models wind as a single complex variable, combining speed and direction, with complex ARIMA (cARIMA). The first experiment (note of 30 September 2026) compared the point forecasts of cARIMA with persistence and with pairs of real ARIMA models, up to 6 hours ahead. This note describes the second experiment, which responds to the comments on that note:

1. **The cAR(0) models looked strange** (most models chosen on the selection window had no AR term). The orders are now chosen by cross-validation on separate forecast origins, and the accuracy of every order is reported on the test origins (Sections 4.2 and 5).
2. **Horizon up to 2 hours**: $h = 1, \ldots, 12$ steps of 10 minutes, for the choice of the orders, the forecasts and the evaluation.
3. **Phase plane**: magnitude against angle for Z_t and $\log Z_t$, and for their changes ΔZ_t and $\Delta \log Z_t$ (Section 2.1).
4. **cARIMA in logarithms with a real constant**: a drift in log speed but none in the direction (Section 3.7).
5. **Predictive distributions**: the quantiles $\tau = 0.01, 0.02, \ldots, 0.99$ of speed and direction for every model, evaluated by the pinball score for each quantile and by calibration plots (Sections 4.4 and 7).

The data and the models are the same as in the first experiment; Sections 2 and 3 repeat their description so that this note can be read on its own.

2 Data

- **Source.** The Kaggle "Wind Turbine SCADA" dataset, file `T1.csv`. SCADA (supervisory control and data acquisition) is the monitoring system of a wind turbine; it records measurements at the turbine, here every 10 minutes. The file covers one turbine in Turkey for the whole of 2018 and contains wind speed (m/s) and wind direction (degrees). More details are in `data/README.md`.
- **Missing values.** A complete 10-minute grid for 2018 has 52,560 time points; the file has 50,530 of them. The 2,030 missing points include three gaps longer than 3 days.
- **Preparation** (`R/02_prepare_data.R`):
  - the data are put on the complete 10-minute grid, with the missing points left empty;
  - gaps of at most 36 points (6 hours) are filled by linear interpolation of the two Cartesian coordinates of the wind vector, $\mathrm{Re}\, Z_t$ and $\mathrm{Im}\, Z_t$ (defined in Section 3.1);

– the 10 zero speeds are replaced by 0.1 m/s, so that the logarithm of speed exists.

The 10-minute changes of wind are heavy-tailed: most changes are small, but some are very large. Table 1 gives the kurtosis $\mathrm{E}(x - \mathrm{E}\,x)^4/(\mathrm{Var}\,x)^2$ of the changes between consecutive observations. It equals 3 for a normal distribution; larger values mean heavier tails. All four changes are far from normal. About 3% of the direction changes exceed 30°, and 0.5% exceed 90°. The changes of the Cartesian coordinates combine changes of speed and of direction, and $\mathrm{Re}\,Z_t$ has the heaviest tails of all: a sudden turn at high speed moves the wind vector a long way.

Figure 1 shows the same changes as histograms with the counts on a log scale, so that the tails are visible; a normal distribution would appear as a parabola.

- **Left, change of log speed.** The distribution is sharply peaked at zero: half of the changes are within ±0.06 (about ±6% of the speed), with a standard deviation of 0.16. But the counts decay roughly linearly on the log scale, i.e. exponentially rather than like a normal distribution, and reach changes of −1.6 and +2.4 (speeds falling to a fifth or rising elevenfold within 10 minutes). Such large relative changes happen at low speeds: in all 66 changes larger than 1 in absolute value, one of the two speeds is below 3.7 m/s.
- **Middle, change of the recorded angle.** Besides the central peak there are two side peaks near $\pm 2\pi$. They are not real turns of the wind but the artificial jumps that occur when the direction crosses $\pm\pi$ (696 times; Section 3.6). Because of them the standard deviation is 0.71 rad.
- **Right, shortest-arc change of direction.** With the jumps removed, the side peaks disappear and the standard deviation falls to 0.25 rad (14°). The centre is again very narrow (half of the changes are within about ±3°), and the tails are heavy and extend to $\pm\pi$: there are 63 turns of more than 150° within 10 minutes.

These heavy tails motivate the heavy-tailed error distribution in Section 3.4, and the side peaks motivate the unwrapping of the angle in Section 3.6.

Table 1: Kurtosis of the changes between consecutive 10-minute observations (normal distribution: 3). v_t is speed, θ_t direction and Z_t the wind as a complex number (Section 3.1); wrap(·) is defined in (9).

change	kurtosis
$\log v_t - \log v_{t-1}$	14.4
$\mathrm{wrap}(\theta_t - \theta_{t-1})$	52.8
$\mathrm{Re}\,Z_t - \mathrm{Re}\,Z_{t-1}$	131.9
$\mathrm{Im}\,Z_t - \mathrm{Im}\,Z_{t-1}$	25.5

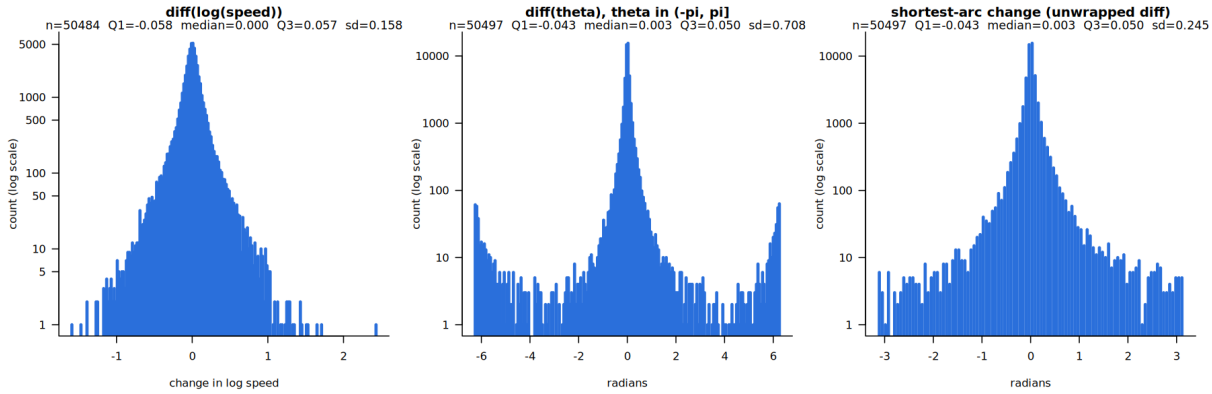


Figure 1: Histograms of the changes between consecutive 10-minute observations, counts on a log scale. Left: change of log speed. Middle: change of the direction angle as recorded in $(-\pi, \pi]$, in radians. Right: shortest-arc change of direction, in radians. The line above each panel gives the number of changes, the quartiles and the standard deviation.

2.1 Phase plane

Figures 2 and 3 plot the magnitude (horizontal axis) against the angle (vertical axis) for all 10-minute observations of 2018 (`R/08_phase_plane.R`). In Figure 2 the angle of $\log Z_t$ uses the recorded direction in $(-\pi, \pi]$.

- Z_t **(Figure 2, left).** Speed against direction. The two prevailing directions, about 60° and −160° to 180°, are horizontal bands, and the strongest winds come from them. Other directions occur mostly at low speeds.
- $\log Z_t$ **(right).** $|\log Z_t| = \sqrt{\log^2 v_t + \theta_t^2}$ against $\operatorname{Arg} \log Z_t = \operatorname{atan2}(\theta_t, \log v_t)$. The two prevailing directions become two tight bands, at about +25° and −50°, which narrow as the speed grows.
- $\Delta Z_t = Z_t - Z_{t-1}$ **(Figure 3, left).** The 10-minute changes of the wind vector go in all directions with no preferred angle; half of them are shorter than 0.62 m/s.
- $\Delta \log Z_t = \log(v_t / v_{t-1}) + \mathrm{i}\, \mathrm{wrap}(\theta_t - \theta_{t-1})$ **(right).** The changes the models with $d = 1$ work on. They concentrate near the angles 0° and ±180°: most 10-minute changes are mainly a change of log speed, with a much smaller change of direction (in radians). The large changes, to the right, are spread over all angles.

Differences are only taken between consecutive observed points (50,707 pairs), and the horizontal axes of Figure 3 are cut at the 99.5% quantile of the modulus (the maxima are 34 m/s and 5.6).

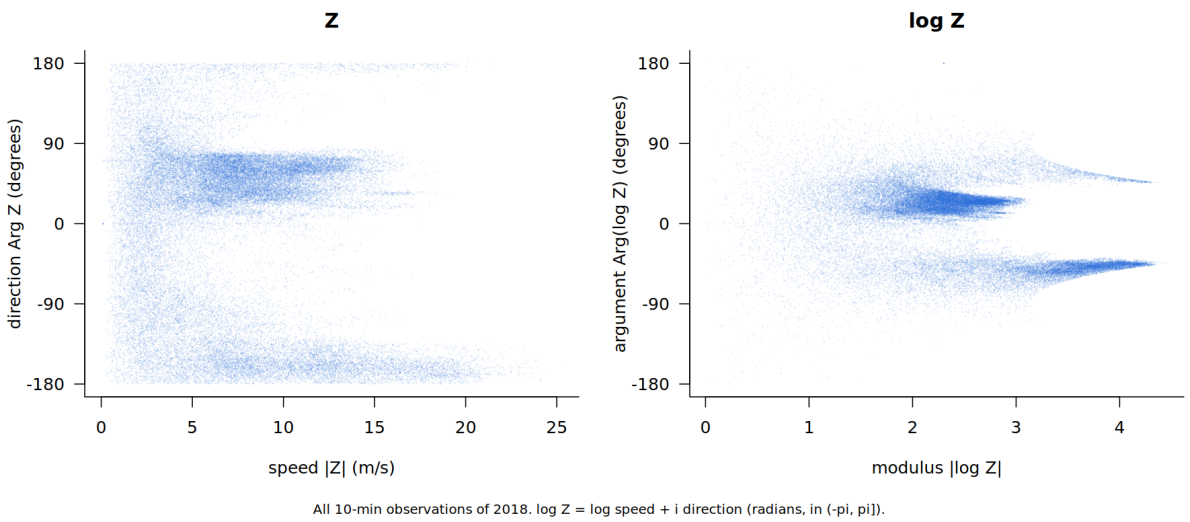


Figure 2: Phase plane of Z_t (speed against direction) and of $\log Z_t$ (modulus against argument), all 10-minute observations of 2018.

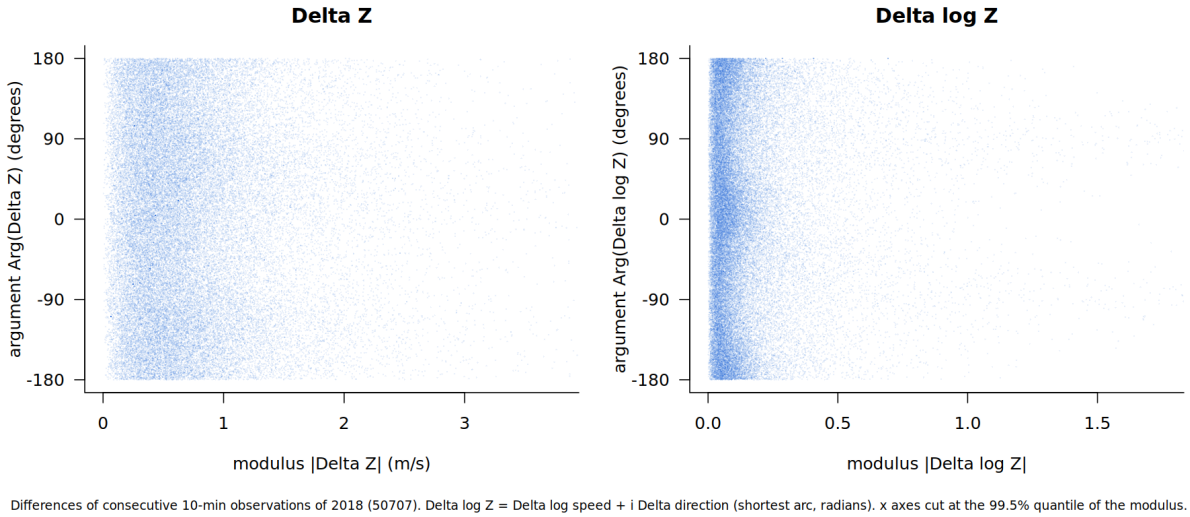


Figure 3: Phase plane of the 10-minute changes ΔZ_t and $\Delta \log Z_t$ (modulus against argument).

3 How the wind is modelled

3.1 Wind as a complex number

With speed $v_t > 0$ and direction angle $\theta_t \in (-\pi, \pi]$ (in radians), the wind at time t is

$$Z_t = v_t e^{\mathrm{i}\theta_t} = v_t \cos\theta_t + \mathrm{i}\, v_t \sin\theta_t. \tag{1}$$

The modulus $|Z_t|$ is the speed and the argument $\operatorname{Arg} Z_t$ is the direction. $\operatorname{Re} Z_t$ and $\operatorname{Im} Z_t$ are the Cartesian coordinates of the wind vector. They are not the zonal and meridional components used in meteorology, which are measured along the east and north axes; here the angle is simply the recorded direction.

3.2 Complex ARIMA

The cARIMA(p, d, q) model for a complex series y_t is

$$\Big(1 - \sum_{j=1}^{p} \phi_j B^j\Big)(1 - B)^d \, y_t = \beta_0 + \Big(1 + \sum_{j=1}^{q} \theta_j B^j\Big)\varepsilon_t, \qquad \phi_j, \theta_j, \beta_0 \in \mathbb{C}, \tag{2}$$

where B is the backshift operator ($By_t = y_{t-1}$), p and q are the autoregressive (AR) and moving average (MA) orders, and d is the order of differencing. With $d = 1$ the constant β_0 is a *drift*: a fixed complex change added at every step. With $d = 0$ it determines the mean of the series, $\beta_0/(1 - \sum_j \phi_j)$.

Multiplying by a complex coefficient $\phi = a + \mathrm{i}b = re^{\mathrm{i}\alpha}$ *scales* by $r = |\phi|$ and *rotates* by $\alpha = \operatorname{Arg}\phi$. For the pair $(\operatorname{Re} y, \operatorname{Im} y)$,

$$\begin{pmatrix} \operatorname{Re}(\phi y) \\ \operatorname{Im}(\phi y) \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} \operatorname{Re} y \\ \operatorname{Im} y \end{pmatrix}, \tag{3}$$

so cARIMA is a bivariate VARIMA whose coefficient matrices are restricted to this form: 2 real parameters per lag instead of 4.

The models are estimated by maximum likelihood with `complex::clm()`. It fits (2) for the series in levels: the observations lost to differencing and lags at the start of the sample are filled in by extrapolation, and the forecasts are produced directly for y_t rather than for its differences.

3.3 The error distribution and non-circularity

In the basic model, the error ε_t is *complex normal*. For a complex random variable with zero mean, two second moments are needed, and we use the terms of Svetunkov and Svetunkov (2024):

- the *conjugate variance* $\sigma^2 = \mathrm{E}(\varepsilon_t \bar{\varepsilon}_t) = \mathrm{E}\,|\varepsilon_t|^2$, computed with the complex conjugate; it is real and measures the overall size of the error;
- the *direct variance* $\varsigma^2 = \mathrm{E}(\varepsilon_t^2)$, computed without the conjugate (elsewhere called the pseudo-variance); it is complex in general and describes the shape of the error cloud in the plane.

Together they give the covariance matrix of the real and imaginary parts:

$$\operatorname{Var}(\operatorname{Re}\varepsilon_t) = \tfrac{1}{2}(\sigma^2 + \operatorname{Re}\varsigma^2), \quad \operatorname{Var}(\operatorname{Im}\varepsilon_t) = \tfrac{1}{2}(\sigma^2 - \operatorname{Re}\varsigma^2), \quad \operatorname{Cov}(\operatorname{Re}\varepsilon_t, \operatorname{Im}\varepsilon_t) = \tfrac{1}{2}\operatorname{Im}\varsigma^2. \tag{4}$$

With $\varsigma^2 = 0$ the error is *circular*: the two parts have the same variance and are uncorrelated, so the cloud of errors is a circle. With $\varsigma^2 \neq 0$ it is *non-circular*: the cloud is an ellipse. For wind, non-circularity means that the changes of the wind vector are larger along some directions than others. We measure it by the ratio of the two variances,

$$\kappa = \frac{|\varsigma^2|}{\sigma^2} \in [0, 1], \tag{5}$$

which is 0 for a circular error and 1 for an error that always lies on one line. A pair of independent real models, one for $\operatorname{Re} Z_t$ and one for $\operatorname{Im} Z_t$, ignores the correlation between the two errors.

3.4 The complex generalised normal distribution

Because the changes of wind are heavy-tailed (Table 1), the models are also estimated with a *complex generalised normal* error (Novey et al., 2010; `distribution = "dcgnorm"` in `clm()`). For the error $e = z - \mu$, with conjugate scale $\alpha^2 > 0$, direct scale $\psi^2 \in \mathbb{C}$ ($|\psi^2| < \alpha^2$) and shape $\beta > 0$, the density is

$$f(z) = \frac{2\beta}{\pi\,\Gamma(1/\beta)\,2^{1/\beta}\sqrt{\alpha^4 - |\psi^2|^2}} \exp\Big(-\tfrac{1}{2}Q^{\beta}\Big), \qquad Q = \frac{2(\alpha^2|e|^2 - \operatorname{Re}(\overline{\psi^2}\, e^2))}{\alpha^4 - |\psi^2|^2}. \tag{6}$$

- $\beta = 1$ gives the complex normal distribution with conjugate variance α^2 and direct variance ψ^2. A shape $\beta < 1$ gives heavier tails ($\beta = 0.5$ is similar to the Laplace distribution); $\beta > 1$ gives lighter tails.
- For other β, the conjugate and direct variances equal α^2 and ψ^2 multiplied by $2^{1/\beta}\Gamma(2/\beta)/(2\Gamma(1/\beta))$. In `dcgnorm()` the two scales are the arguments `scaleConjugate` and `scaleDirect`.
- The shape β is estimated together with the ARIMA coefficients. For any given values of these, the maximum likelihood estimates of the two scales have a closed form, so they are substituted into the likelihood ("concentrated out") and not searched for numerically.

With heavy tails, a few very large changes can pull the estimates of a normal model, in particular the drift. A heavy-tailed likelihood gives these observations less weight.

3.5 Why logarithms

Besides Z_t itself, the models are applied to the complex logarithm of the wind,

$$\log Z_t = \log v_t + \mathrm{i}\,\tilde{\theta}_t, \tag{7}$$

where $\tilde{\theta}_t$ is the unwrapped direction angle (Section 3.6).

- The logarithm *separates speed and direction*: the real part is log speed and the imaginary part is the angle. A cARIMA for $\log Z_t$ is still one complex model: the two parts are linked through the complex coefficients and the error covariance.
- Changes of speed are roughly *proportional* to the speed (larger when it is windy), so log speed is a more natural scale than speed.
- Forecasts are transformed back by $\hat{Z} = \exp(\widehat{\log Z})$, which always gives a positive speed and a valid direction.
- $\exp(\widehat{\log Z})$ is not the expected value of Z; no correction for this is made.
- The drift of a model for $\log Z_t$ with $d = 1$ has a direct meaning: $\mathrm{Re}\,\beta_0$ is a steady percentage change of speed, and $\mathrm{Im}\,\beta_0$ is a steady rotation of the direction, in radians per 10 minutes. There is no physical reason for the wind to keep turning in one direction for weeks, so models for $\log Z_t$ are also fitted without the constant, and with the real part of the constant only (Section 3.7).

3.6 Differences and unwrapping of the angle

- At 10-minute resolution the wind is very persistent: over several weeks its level wanders without returning to a stable mean. The models are therefore fitted mainly to changes, $d = 1$. In logarithms the differences have a clear meaning:

$$(1 - B)\log Z_t = \log\frac{v_t}{v_{t-1}} + \mathrm{i}\,(\tilde{\theta}_t - \tilde{\theta}_{t-1}), \tag{8}$$

  the log ratio of consecutive speeds plus i times the change of direction.
- The recorded angle jumps by about 2π whenever the direction crosses $\pm\pi$. For example, a turn from 179° to −179° is a change of 2°, but the recorded values differ by 358°. This happens 696 times in `T1.csv`. In differences these jumps would appear as huge artificial shocks (Figure 1, middle).
- The angle is therefore *unwrapped* within each estimation window $t = 1, \ldots, T$:

$$\tilde{\theta}_1 = \theta_1, \qquad \tilde{\theta}_t = \tilde{\theta}_{t-1} + \mathrm{wrap}(\theta_t - \theta_{t-1}), \qquad \mathrm{wrap}(x) = \mathrm{atan2}(\sin x, \cos x) \in (-\pi, \pi]. \tag{9}$$

wrap$(\cdot)$ returns the equivalent angle in $(-\pi, \pi]$, so each step of $\tilde{\theta}_t$ is the *shortest-arc* change of direction, and $\tilde{\theta}_t$ can leave $(-\pi, \pi]$. Forecasts of the direction are wrapped back into $(-\pi, \pi]$ for evaluation.

- The unwrapped angle is used in $\log Z_t$ and in the real benchmark models for the angle (Section 4.2). Z_t itself needs no unwrapping: it is continuous when the direction crosses $\pm\pi$, because cos and sin are periodic.
- For Z_t, $d = 0$ is also a plausible choice: the wind vector could revert to a mean. One set of models chooses $d \in \{0, 1\}$ automatically (Section 4.2).

3.7 The constant

With the partial intercept of `clm()`, the constant β_0 of (2) can be complex, real, imaginary or absent (in the formula: `y ~ 1`, `y ~ 1 - Im(1)`, `y ~ 1 - Re(1)`, `y ~ -1`). For $\log Z_t$ with $d = 1$, a real constant is a steady percentage change of speed with no rotation of the direction. In the first experiment, the complex constant of the normal model rotated the direction forecasts by about 0.7° per 10 minutes and made them worse. Three constants are therefore compared for $\log Z_t$: complex, real only (suffix `_re`) and none (suffix `_nd`).

4 Design of the forecasting study

4.1 Rolling origin

As in the first experiment, the models are repeatedly estimated on a window of past data and forecast the next 2 hours, for many different ends of the window.

- The estimation window is the 4 weeks up to the forecast origin ($T = 4032$ observations), and the horizon is $h = 1, \ldots, 12$ steps ahead (10 minutes to 2 hours).
- The *test origins* are the same 300 origins as in the first experiment, from 2018-02-27 14:40 to 2018-12-31 17:50 (all of them remain usable with the shorter horizon). The results are therefore directly comparable with the first 12 horizons of the first experiment; for persistence they are identical.
- At every origin, the parameters of all models are estimated again on that origin's window; only the orders stay fixed.

4.2 Choice of model orders

- **cARIMA with $d = 1$: cross-validation.** 100 *cross-validation origins* are spread evenly over the usable origins from 2018-02-27 16:40 to 2018-12-31 21:50 (13–15 in each month from March to September except June, which has 1; 5 in November, 9 in December and 1 at the end of February). Each is at least 12 steps away from every test origin, so no cross-validation forecast covers the same 2 hours as a test forecast; their estimation windows do overlap the test windows, as there is no earlier data. At every cross-validation origin, all 16 models with $p, q \in \{0, \ldots, 3\}$ are estimated and forecast; the orders with the lowest RMSE of the wind vector over the 100 origins and $h = 1, \ldots, 12$ (Section 4.5) are chosen, separately for each variant (series, constant and distribution).
- **Diagnostic on the test origins.** For the two models for $\log Z_t$ without a constant (normal and generalised normal), all 16 orders are also forecast at the 300 test origins. This shows how much the choice of p and q matters on the data used for the evaluation. It is not used to choose anything.

- **Starting values.** Every cARIMA model is estimated from two starting values, and the fit with the highest likelihood is kept: the default of `clm()` (the complex least squares estimates for the differenced series, with 0.1+0.1i for each MA coefficient) and the Hannan–Rissanen estimates. The Hannan–Rissanen method regresses the series on its own lags and on the lags of errors approximated by a long autoregression, by complex least squares; it gives consistent estimates of the AR and MA coefficients without numerical optimisation.
- **`auto.carima()` models and real benchmarks.** Their orders (and d and the constant of `auto.carima()`) are those chosen by AICc on the selection window in the first experiment: `auto.carima()` with a stepwise search over $p, q \leq 3$ and $d \in \{0, 1\}$; `forecast::auto.arima()` with $d = 1$; `smooth::auto.msarima()` on the differenced series, refitted with $d = 1$.

4.3 Models compared

Table 2 lists the models. The names are built as in the first experiment:

- `carima_`: cARIMA with $d = 1$ and orders from cross-validation; `auto_carima_`: cARIMA with d, the orders and the constant chosen by `auto.carima()`;
- `logZ` or `z`: the model is for $\log Z_t$ or for Z_t;
- `_cgn`: complex generalised normal error (otherwise complex normal);
- `_nd`: no constant; `_re`: real constant only; otherwise a complex constant;
- `arima_` and `msarima_`: two independent real ARIMA models, from `auto.arima()` and `auto.msarima()` respectively, for the pair of series named after the prefix. Their forecasts are combined into $\hat{Z}$: `speed_angle` (speed and unwrapped angle), `logSpeed_angle` (log speed and unwrapped angle) and `re_im` ($\mathrm{Re}\, Z_t$ and $\mathrm{Im}\, Z_t$). A negative speed forecast is set to zero.

Table 2: Models compared, with the orders used for all test origins. For the `carima_` models the orders come from cross-validation; the orders chosen by AICc in the first experiment are given for comparison.

model	series	error	constant β_0	(p, d, q)	first experiment
`persistence`	–	empirical	–	$\hat{Z}_{t+h\|t} = Z_t$	
`carima_logZ`	$\log Z$	normal	complex	(3,1,1)	(0,1,3)
`carima_logZ_cgn`	$\log Z$	gen. normal	complex	(1,1,3)	(0,1,2)
`carima_logZ_nd`	$\log Z$	normal	none	(2,1,1)	(1,1,2)
`carima_logZ_cgn_nd`	$\log Z$	gen. normal	none	(3,1,0)	(0,1,2)
`carima_logZ_re`	$\log Z$	normal	real	(0,1,2)	(1,1,1)
`carima_logZ_cgn_re`	$\log Z$	gen. normal	real	(3,1,2)	(0,1,2)
`carima_z`	Z	normal	complex	(1,1,1)	(0,1,2)
`carima_z_cgn`	Z	gen. normal	complex	(3,1,3)	(0,1,2)
`auto_carima_logZ`	$\log Z$	normal	complex	(1,1,1)	same
`auto_carima_logZ_cgn`	$\log Z$	gen. normal	none	(0,1,2)	same
`auto_carima_z`	Z	normal	complex	(1,0,2)	same
`auto_carima_z_cgn`	Z	gen. normal	complex	(1,0,2)	same
`arima_*`, `msarima_*`	two real series	normal	chosen	$d = 1$, see note of 30 Sep.	same

4.4 Predictive distributions

Every model gives, at every origin and horizon, a point forecast $\hat{Z}_{t+h|t}$ and the quantiles $\tau = 0.01, 0.02, \ldots, 0.99$ of speed $|Z_{t+h}|$ and of direction $\mathrm{Arg}\, Z_{t+h}$. Parameter uncertainty is ignored throughout.

cARIMA. The forecast error of (2) h steps ahead is a weighted sum of the future innovations,

$$y_{t+h} - \hat{y}_{t+h|t} = \sum_{j=0}^{h-1} \psi_j \, \varepsilon_{t+h-j}, \qquad \psi_0 = 1, \quad \psi_j = \theta_j + \sum_{i=1}^{\min(j,\, p+d)} a_i \, \psi_{j-i}, \tag{10}$$

where a_i are the coefficients of the AR and differencing polynomial, $(1 - \sum_j \phi_j B^j)(1 - B)^d = 1 - \sum_i a_i B^i$, and $\theta_j = 0$ for $j > q$. Its conjugate and direct variances are $\sigma^2 \sum_j |\psi_j|^2$ and $\varsigma^2 \sum_j \psi_j^2$. These are implemented in `forecast()` for `carima` objects and `predict()` for `clm` objects in the `complex` package (development version; the arguments follow `smooth::forecast.adam()`):

- `interval = "approximate"`: the real and imaginary parts are taken as normal with the variances and covariance of Section 3.3; this is exact for the complex normal, and is what `interval = "prediction"` uses for it;
- `interval = "simulated"`: `nsim` paths of innovations are drawn from the estimated error distribution and combined with (10). `interval = "prediction"` uses this for the generalised normal, because a sum of generalised normal variables is not generalised normal.

On 300 simulated cARIMA(1,1,1) series, the 80% and 95% intervals covered 78.5% and 95.0% of the future values with the normal error, and 80.4% and 95.9% with the generalised normal.

From the distribution of y_{t+h} to speed and direction:

- $\log Z_t$**:** speed $= \exp(\mathrm{Re} \log Z)$ and direction $= \mathrm{Im} \log Z$ are each a monotone function of one part, so their quantiles are $\exp(\cdot)$ of the quantiles of the real part and the quantiles of the imaginary part: from the normal distribution for the normal models, and from 10,000 simulated paths for the generalised normal.
- Z_t**:** speed $|Z|$ and direction $\mathrm{Arg}\, Z$ depend on both parts, so their quantiles are taken from 10,000 values drawn from the joint distribution at each horizon (the complex normal with the variances above, or the simulated paths). The directions are measured relative to the direction of the point forecast, so they have no jump at $\pm\pi$.

The point forecast is the conditional mean of y_{t+h} (for $\log Z$ transformed back by $\exp(\cdot)$, as before), not a quantile of the distribution.

Real pairs. Each real ARIMA model gives a normal predictive distribution (its standard deviation from the interval of ± 1 standard deviation of `forecast::forecast()` or `smooth`'s `forecast()`). Speed comes from the speed model (with negative quantiles set to zero) or as $\exp(\cdot)$ of the quantiles of log speed, and direction from the angle model. For `re_im`, speed and direction are taken from 10,000 values drawn from the two independent normal distributions at each horizon.

Persistence has no model, so its distribution is a bootstrap of its own errors in the estimation window: the quantiles of speed and direction at $t + h$ are

$$v_t + Q_\tau\{v_{s+h} - v_s\}, \qquad \theta_t + Q_\tau\{\mathrm{wrap}(\theta_{s+h} - \theta_s)\}, \tag{11}$$

where Q_τ is the empirical τ-quantile over all s in the window with $s + h \le t$. Negative speeds are set to zero.

4.5 Error measures

Point forecasts. As in the first experiment, for origin t and horizon h:

$$e_{t,h}^{\text{speed}} = |\hat{Z}_{t+h|t}| - |Z_{t+h}| \quad \text{(m/s)}, \tag{12}$$

$$e_{t,h}^{\text{angle}} = \tfrac{180}{\pi}\, \mathrm{wrap}(\mathrm{Arg}\, \hat{Z}_{t+h|t} - \mathrm{Arg}\, Z_{t+h}) \quad \text{(degrees)}, \tag{13}$$

$$e_{t,h}^{\text{vector}} = |\hat{Z}_{t+h|t} - Z_{t+h}| \quad \text{(m/s)}, \tag{14}$$

each summarised by the RMSE over the 300 origins, for each horizon and over all 12 horizons together (column "$h = 1$–12").

Predictive distributions, for speed and for direction separately. With $q_{t,h,\tau}$ the forecast τ-quantile and x_{t+h} the actual value:

- the *pinball score* (quantile loss)

$$\rho_\tau(x, q) = (x - q)\,(\tau - \mathbf{1}\{x < q\}), \tag{15}$$

  averaged over the 300 origins for each τ and h. Lower is better. Its average over τ approximates half of the continuous ranked probability score (CRPS), and is in the units of the target (m/s or degrees);

- the *calibration*

$$c_{h,\tau} = \frac{1}{300} \sum_t \mathbf{1}\{x_{t+h} \le q_{t,h,\tau}\} - \tau, \tag{16}$$

  the share of actual values below the forecast τ-quantile minus its nominal value. A perfectly calibrated forecast has $c_{h,\tau} = 0$ for all τ. Calibration plots show $c_{h,\tau}$ (vertical axis) against τ (horizontal axis). As a summary we report the mean of $|c_{h,\tau}|$ over τ and the coverage of the central 50%, 80% and 98% intervals (from the quantiles 0.25–0.75, 0.1–0.9 and 0.01–0.99).

Direction is circular. Each actual direction is therefore taken as the representative $\operatorname{Arg} Z_{t+h} + 2\pi k$ closest to the forecast median ($\tau = 0.5$), on the same unwrapped scale as the quantiles.

5 Model orders

Table 3 summarises the cross-validation: for each variant, the chosen orders and their vector RMSE, the best orders with $p = 0$, and the spread between the best and the worst of the 16 orders. All 25,600 fits at the cross-validation origins succeeded.

- $p > 0$ **is chosen for 7 of the 8 variants**, so cross-validation does not support cAR(0). Only `carima_logZ_re` has $p = 0$.
- **But the orders hardly matter.** For every variant, all 16 orders are within 0.4–1.0% of each other, and the best model with $p = 0$ is at most 0.6% worse than the best model.
- This agrees with a diagnostic on 8 windows made before this experiment: in fits of cARIMA$(1, 1, 1)$ the AR and MA coefficients nearly cancel, $|\hat{\phi}_1 + \hat{\theta}_1| = 0.03$–$0.12$, so $(1 - \phi_1 B)$ and $(1 + \theta_1 B)$ are almost a common factor and the model is close to cARIMA$(0, 1, 0)$. The orders are weakly identified: different windows and criteria choose between $(0, 1, q)$, $(p, 1, 0)$ and $(1, 1, 1)$ almost at random, with little effect on the forecasts. Part of the cAR(0) in the first experiment came from this, part from a sub-optimal fit on the selection window (the normal model for $\log Z$ without a constant prefers $(1, 1, 1)$ there by 10.7 AICc once the optimiser also starts from the Hannan–Rissanen estimates).

Table 3: Cross-validation: RMSE of the wind vector (m/s) over 100 origins and $h = 1$–12. Persistence: 2.147.

variant	chosen	RMSE	best with $p = 0$	RMSE	best to worst
`carima_logZ`	(3,1,1)	2.1950	(0,1,2)	2.2000	0.50%
`carima_logZ_cgn`	(1,1,3)	2.1298	(0,1,3)	2.1355	0.72%
`carima_logZ_nd`	(2,1,1)	2.1404	(0,1,2)	2.1420	0.37%
`carima_logZ_cgn_nd`	(3,1,0)	2.1368	(0,1,3)	2.1371	0.48%
`carima_logZ_re`	(0,1,2)	2.1397	(0,1,2)	2.1397	0.64%
`carima_logZ_cgn_re`	(3,1,2)	2.1257	(0,1,2)	2.1321	0.92%
`carima_z`	(1,1,1)	2.1365	(0,1,1)	2.1381	0.82%
`carima_z_cgn`	(3,1,3)	2.1217	(0,1,3)	2.1350	1.00%

The diagnostic on the 300 test origins (Table 4, Figure 4) tells the same story. For both models for $\log Z$ without a constant, the vector RMSE of all 16 orders lies within 1.1% (generalised normal: 0.8%). The best order with $p > 0$ is better than the best with $p = 0$ by only 0.1–0.2%. The orders chosen by cross-validation are among the best on the test origins: $(2, 1, 1)$ is the best of all for the normal model (2.3002), and $(3, 1, 0)$ is 0.2% behind the best, $(2, 1, 2)$, for the generalised normal (2.2949 against 2.2901).

Table 4: Diagnostic on the 300 test origins, $h = 1$–12: for each p, the q with the lowest vector RMSE and its RMSE of the vector (m/s), speed (m/s) and direction (degrees).

model	p	best q	vector	speed	direction
`carima_logZ_nd`	0	2	2.3034	1.5611	27.10
	1	0	2.3049	1.5563	27.35
	2	1	**2.3002**	1.5605	27.14
	3	0	2.3023	1.5595	27.13
`carima_logZ_cgn_nd`	0	3	2.2936	1.5532	27.43
	1	2	2.2931	1.5550	27.40
	2	2	**2.2901**	1.5570	27.41
	3	3	2.2925	1.5537	27.37

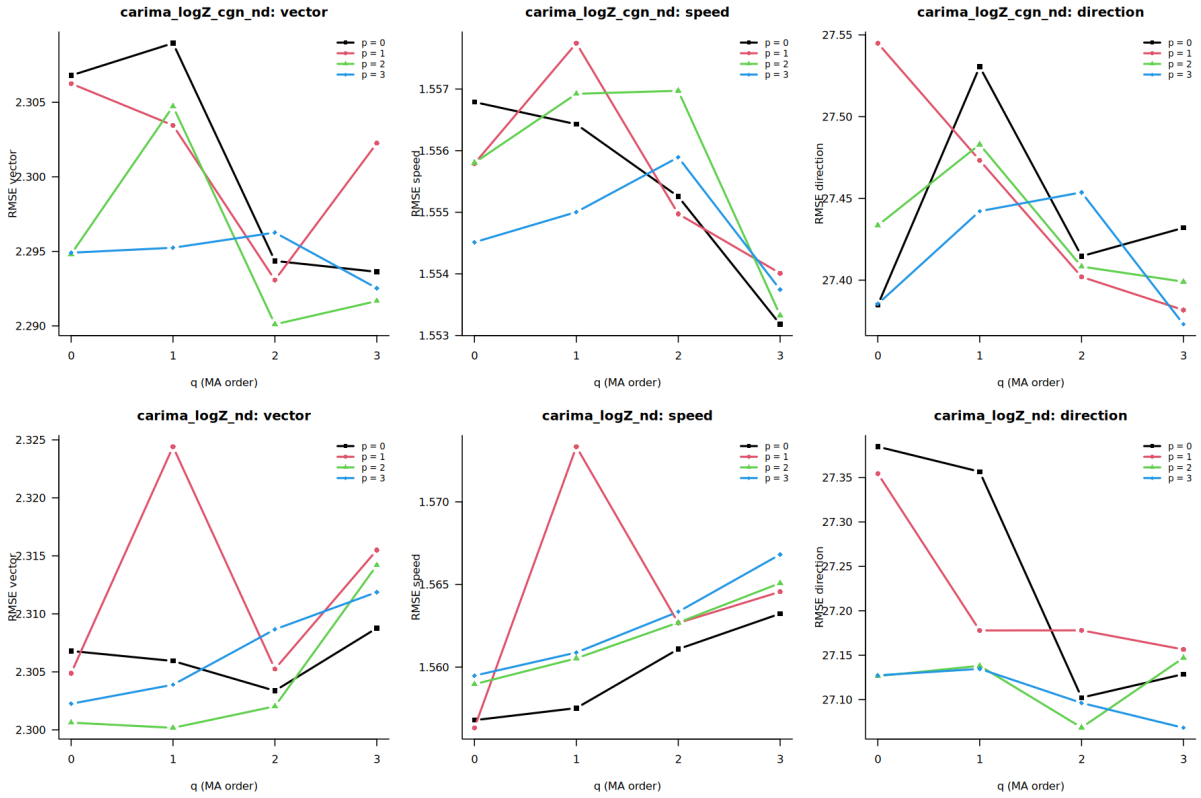


Figure 4: RMSE over the 300 test origins and $h = 1$–12 of all cARIMA$(p, 1, q)$ for $\log Z$ without a constant (top: generalised normal, bottom: normal), against q, one line for each p.

6 Accuracy of the point forecasts

Tables 5–7 give the RMSE over the 300 test origins at selected horizons; all 12 horizons are in `results/exp2/tables/rmse_{speed,direction,vector}.csv`. Bold marks the lowest value in each column. All models produced forecasts at all origins. The differences between models are small: the best model in each table is less than 1% better than persistence over all horizons.

- **Speed.** At $h = 1$ and $h = 3$ the cARIMA models for $\log Z$ with the generalised normal error are best (0.705 and 1.044 m/s against 0.710 and 1.048 for persistence), as in the first experiment. From $h = 6$ on, `arima_speed_angle`, which models speed directly, is best (1.545 over all horizons against 1.557). The models for Z with $d = 0$ (`auto_carima_z`, `auto_carima_z_cgn`) are already worse than persistence at 2 hours, because their forecasts move towards the short mean wind vector.
- **Direction.** The normal models for $\log Z$ without a rotating drift are best: `carima_logZ_re` (27.10°) and `carima_logZ_nd` (27.14°) against 27.38° for persistence. The models with a rotating drift (`auto_carima_logZ` and the real pairs with the angle model) are the worst, but over 2 hours the rotation has much less effect than over 6 hours.
- **Vector.** `auto_carima_z_cgn` (2.291), `carima_z_cgn` (2.293) and the cARIMA models for $\log Z$ with the generalised normal error (2.294–2.300) are best, against 2.307 for persistence; `carima_logZ_cgn_nd` is best at $h = 1$. The real pairs with the angle model are 4–5% worse than persistence.

Table 5: RMSE of speed $|\hat{Z}|$ (m/s).

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 1$–12
`persistence`	0.710	1.048	1.571	1.995	1.557
`carima_logZ`	0.714	1.055	1.570	1.994	1.560
`carima_logZ_cgn`	0.707	1.046	1.577	2.014	1.565
`carima_logZ_nd`	0.713	1.054	1.571	1.994	1.560
`carima_logZ_cgn_nd`	0.706	1.046	1.568	1.990	1.554
`carima_logZ_re`	0.713	1.059	1.575	2.001	1.565
`carima_logZ_cgn_re`	0.706	1.050	1.574	2.005	1.563
`carima_z`	0.716	1.054	1.569	1.990	1.556
`carima_z_cgn`	0.717	1.052	1.565	1.989	1.555
`auto_carima_logZ`	0.711	1.058	1.571	1.980	1.558
`auto_carima_logZ_cgn`	**0.705**	**1.044**	1.569	1.992	1.555
`auto_carima_z`	0.724	1.074	1.601	2.059	1.607
`auto_carima_z_cgn`	0.721	1.066	1.589	2.042	1.591
`arima_logSpeed_angle`	0.727	1.083	1.598	2.014	1.592
`arima_re_im`	0.727	1.054	1.569	1.978	1.557
`arima_speed_angle`	0.720	1.047	**1.557**	**1.951**	**1.545**
`msarima_logSpeed_angle`	0.731	1.089	1.609	2.019	1.598
`msarima_re_im`	0.728	1.061	1.573	1.996	1.563
`msarima_speed_angle`	0.718	1.052	1.571	1.972	1.558

Table 6: RMSE of direction (degrees). The pairs `*_speed_angle` and `*_logSpeed_angle` use the same angle model and have identical rows.

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 1$–12
`persistence`	12.42	20.74	26.78	36.49	27.38
`carima_logZ`	12.66	20.72	26.72	36.96	27.53
`carima_logZ_cgn`	12.51	20.94	26.84	36.57	27.45
`carima_logZ_nd`	12.69	20.55	26.47	36.35	27.14
`carima_logZ_cgn_nd`	12.47	20.83	26.77	36.53	27.39
`carima_logZ_re`	12.70	**20.53**	**26.42**	36.33	**27.10**
`carima_logZ_cgn_re`	12.51	20.94	26.85	36.59	27.44
`carima_z`	12.55	20.93	26.86	36.51	27.45
`carima_z_cgn`	12.73	20.96	26.98	36.34	27.45
`auto_carima_logZ`	12.47	20.95	27.07	37.31	27.87
`auto_carima_logZ_cgn`	12.48	20.85	26.80	36.54	27.41
`auto_carima_z`	12.94	21.43	27.48	36.41	27.70
`auto_carima_z_cgn`	12.76	21.35	27.39	**36.12**	27.58
`arima_logSpeed_angle`	12.49	21.09	27.15	37.27	27.88
`arima_re_im`	12.87	21.14	27.06	36.75	27.63
`arima_speed_angle`	12.49	21.09	27.15	37.27	27.88
`msarima_logSpeed_angle`	**12.38**	21.15	27.29	37.27	27.94
`msarima_re_im`	12.74	20.85	26.78	36.58	27.43
`msarima_speed_angle`	**12.38**	21.15	27.29	37.27	27.94

Table 7: RMSE of the wind vector $|\hat{Z} - Z|$ (m/s).

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 1$–12
`persistence`	0.960	1.634	2.307	2.996	2.307
`carima_logZ`	0.964	1.652	2.356	3.128	2.384
`carima_logZ_cgn`	0.954	1.619	2.300	2.994	2.300
`carima_logZ_nd`	0.961	1.627	2.294	2.989	2.300
`carima_logZ_cgn_nd`	**0.953**	1.622	2.295	2.982	2.295
`carima_logZ_re`	0.962	1.633	2.298	2.998	2.307
`carima_logZ_cgn_re`	0.955	1.625	2.297	2.986	2.298
`carima_z`	0.964	1.639	2.314	2.994	2.311
`carima_z_cgn`	0.965	1.627	2.296	2.971	2.293
`auto_carima_logZ`	0.960	1.657	2.355	3.110	2.379
`auto_carima_logZ_cgn`	0.954	**1.619**	2.294	2.982	2.294
`auto_carima_z`	0.965	1.630	2.300	2.956	2.296
`auto_carima_z_cgn`	0.964	1.636	2.301	**2.951**	**2.291**
`arima_logSpeed_angle`	0.971	1.684	2.390	3.150	2.417
`arima_re_im`	0.974	1.631	**2.294**	2.969	2.298
`arima_speed_angle`	0.968	1.669	2.374	3.132	2.400
`msarima_logSpeed_angle`	0.975	1.696	2.411	3.165	2.432
`msarima_re_im`	0.973	1.645	2.309	2.996	2.312
`msarima_speed_angle`	0.967	1.676	2.383	3.143	2.408

7 Accuracy of the predictive distributions

7.1 Pinball score

Tables 8 and 9 give the pinball score averaged over the 99 quantiles; the score for each τ is in `results/exp2/tables/pinball_by_tau_{speed,direction}.csv`, and Figure 5 plots the scores against the horizon.

- **Direction: cARIMA for Z is best.** `carima_z_cgn` has 5.53 over all horizons against 5.92 for persistence, 7% better, and is best at every horizon except $h = 1$, where `auto_carima_z_cgn` is best. The other models for Z and the pairs of real models for $\operatorname{Re} Z$ and $\operatorname{Im} Z$ (5.58–5.62) follow. The generalised normal improves on the normal error for every series.
- **Speed: no clear winner.** `arima_speed_angle` (0.409), persistence (0.410), `arima_re_im` (0.411), `msarima_speed_angle` (0.412) and `carima_z_cgn` (0.412) are within 1%. `carima_z_cgn` is best at $h = 1$–3 and $h = 5$, `arima_speed_angle` from $h = 7$ on.
- **The models for $\log Z$ have poor distributions** although their point forecasts are among the best. All cARIMA models for $\log Z$ are worse than persistence: by 13–14% for speed and 6% for direction with the generalised normal error, and by 21–24% and 11–15% with the normal error. The pairs of real models with log speed are 14% worse for speed, and the pairs with the angle model are 16–17% worse for direction.

The reason for the poor scores is the calibration, below: the point forecasts of these models are as good as the others', so their quantiles must be too far from the actual values.

Table 8: Pinball score of speed (m/s), averaged over $\tau = 0.01, \ldots, 0.99$.

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 1$–12
`persistence`	0.1908	0.2880	0.4210	0.5474	0.4099
`carima_logZ`	0.2264	0.3595	0.5071	0.6763	0.4980
`carima_logZ_cgn`	0.1973	0.3205	0.4757	0.6454	0.4657
`carima_logZ_nd`	0.2261	0.3592	0.5085	0.6780	0.4991
`carima_logZ_cgn_nd`	0.1971	0.3208	0.4751	0.6433	0.4650
`carima_logZ_re`	0.2261	0.3595	0.5082	0.6774	0.4989
`carima_logZ_cgn_re`	0.1975	0.3216	0.4754	0.6434	0.4654
`carima_z`	0.1924	0.2918	0.4263	0.5697	0.4194
`carima_z_cgn`	**0.1908**	**0.2872**	0.4191	0.5570	0.4123
`auto_carima_logZ`	0.2255	0.3627	0.5169	0.6905	0.5066
`auto_carima_logZ_cgn`	0.1967	0.3201	0.4750	0.6441	0.4651
`auto_carima_z`	0.1943	0.2928	0.4232	0.5516	0.4162
`auto_carima_z_cgn`	0.1915	0.2896	0.4215	0.5538	0.4143
`arima_logSpeed_angle`	0.2280	0.3514	0.4788	0.6067	0.4675
`arima_re_im`	0.1950	0.2894	**0.4180**	0.5473	0.4107
`arima_speed_angle`	0.1947	0.2898	0.4180	**0.5383**	**0.4085**
`msarima_logSpeed_angle`	0.2285	0.3517	0.4812	0.6069	0.4686
`msarima_re_im`	0.1950	0.2918	0.4235	0.5615	0.4166
`msarima_speed_angle`	0.1943	0.2915	0.4209	0.5444	0.4115

Table 9: Pinball score of direction (degrees), averaged over $\tau = 0.01, \ldots, 0.99$.

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 1$–12
`persistence`	2.493	4.429	5.906	8.487	5.917
`carima_logZ`	2.897	4.989	6.642	9.516	6.664
`carima_logZ_cgn`	2.511	4.607	6.297	9.070	6.273
`carima_logZ_nd`	2.903	4.956	6.588	9.349	6.590
`carima_logZ_cgn_nd`	2.507	4.595	6.297	9.068	6.272
`carima_logZ_re`	2.905	4.952	6.571	9.336	6.580
`carima_logZ_cgn_re`	2.519	4.609	6.295	9.072	6.275
`carima_z`	2.341	4.159	5.623	7.946	5.582
`carima_z_cgn`	2.326	**4.112**	**5.571**	**7.848**	**5.525**
`auto_carima_logZ`	2.885	5.046	6.762	9.689	6.775
`auto_carima_logZ_cgn`	2.507	4.596	6.289	9.062	6.270
`auto_carima_z`	2.369	4.221	5.713	7.940	5.622
`auto_carima_z_cgn`	**2.318**	4.161	5.659	7.857	5.570
`arima_logSpeed_angle`	2.874	5.108	6.882	9.823	6.872
`arima_re_im`	2.386	4.209	5.633	7.942	5.610
`arima_speed_angle`	2.874	5.108	6.882	9.823	6.872
`msarima_logSpeed_angle`	2.866	5.130	6.961	9.910	6.929
`msarima_re_im`	2.369	4.170	5.609	7.946	5.585
`msarima_speed_angle`	2.866	5.130	6.961	9.910	6.929

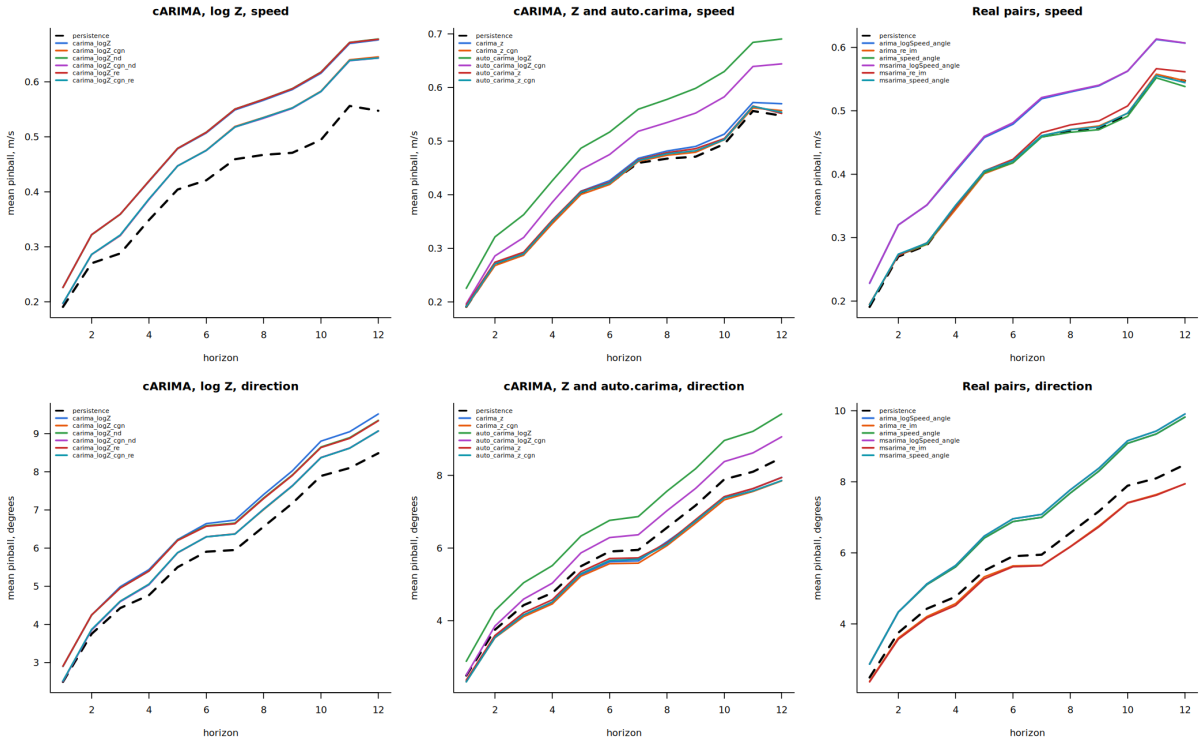


Figure 5: Pinball score averaged over τ, against the horizon. Top: speed (m/s); bottom: direction (degrees). Persistence is the dashed black line in every panel.

7.2 Calibration

Figures 6 and 7 are the calibration plots, for the average over $h = 1$–12 (top) and for $h = 1$ (bottom); Table 10 summarises them. All values for $h = 1$–12, $h = 1$, $h = 6$ and $h = 12$ are in `results/exp2/tables/calibration_{speed,direction}.csv`.

- **Persistence is well calibrated**, with a mean absolute calibration error of 0.014 for speed and 0.026 for direction, and central intervals that cover close to their nominal 50%, 80% and 98%. Its bootstrap reproduces the distribution of the h-step changes in the window.
- **Almost all models have intervals that are too wide in the centre.** The calibration curves are S-shaped: below $\tau = 0.5$ fewer actual values than nominal fall below the quantile, and above $\tau = 0.5$ more. So the quantiles near the median are too far from it. The 50% intervals of the normal models for $\log Z$ cover 72–73% of the actual speeds and 79–80% of the directions; with the generalised normal error, 62–63% and 72%. In the tails it is the other way round: the 98% intervals of most models cover only 96–98%.
- **The best calibrated models** are `auto_carima_z_cgn` for speed (0.015, as good as persistence) and `carima_z_cgn` for direction (0.022, better than persistence). For direction they are followed by the real pairs for $\mathrm{Re}\, Z$ and $\mathrm{Im}\, Z$ and the other models for Z (0.028–0.038). The worst are the normal models for $\log Z$ and the real pairs with the angle model (0.11 for direction).
- **The generalised normal helps most at** $h = 1$: for the models for $\log Z$ it cuts the calibration error of direction from 0.12 to 0.03 at $h = 1$, but only from 0.10 to 0.085 at $h = 12$. The sum (10) of independent generalised normal innovations becomes closer to normal as h grows, while the actual 2-hour changes remain sharply peaked.

Table 10: Calibration over $h = 1$–12: mean absolute calibration error over τ (MACE) and coverage of the central 50%, 80% and 98% intervals.

	speed				direction			
model	MACE	50%	80%	98%	MACE	50%	80%	98%
`persistence`	**0.014**	0.501	0.789	0.978	0.026	0.497	0.794	0.977
`carima_logZ`	0.074	0.722	0.889	0.974	0.108	0.794	0.911	0.968
`carima_logZ_cgn`	0.041	0.624	0.847	0.979	0.077	0.718	0.890	0.971
`carima_logZ_nd`	0.075	0.724	0.890	0.976	0.107	0.799	0.913	0.969
`carima_logZ_cgn_nd`	0.042	0.627	0.849	0.979	0.076	0.716	0.888	0.972
`carima_logZ_re`	0.074	0.723	0.889	0.975	0.106	0.799	0.912	0.968
`carima_logZ_cgn_re`	0.041	0.620	0.849	0.979	0.076	0.716	0.888	0.971
`carima_z`	0.048	0.592	0.852	0.971	0.037	0.605	0.859	0.967
`carima_z_cgn`	0.036	0.556	0.831	0.972	**0.022**	0.556	0.831	0.967
`auto_carima_logZ`	0.077	0.731	0.891	0.977	0.109	0.797	0.912	0.969
`auto_carima_logZ_cgn`	0.041	0.626	0.847	0.981	0.076	0.716	0.889	0.971
`auto_carima_z`	0.024	0.580	0.832	0.967	0.038	0.599	0.852	0.968
`auto_carima_z_cgn`	0.015	0.551	0.821	0.974	0.028	0.568	0.841	0.968
`arima_logSpeed_angle`	0.053	0.652	0.852	0.963	0.114	0.809	0.916	0.972
`arima_re_im`	0.036	0.576	0.837	0.965	0.031	0.587	0.841	0.966
`arima_speed_angle`	0.026	0.579	0.831	0.965	0.114	0.809	0.916	0.972
`msarima_logSpeed_angle`	0.052	0.654	0.853	0.962	0.116	0.815	0.917	0.972
`msarima_re_im`	0.043	0.582	0.841	0.968	0.037	0.603	0.853	0.965
`msarima_speed_angle`	0.027	0.582	0.836	0.963	0.116	0.815	0.917	0.972

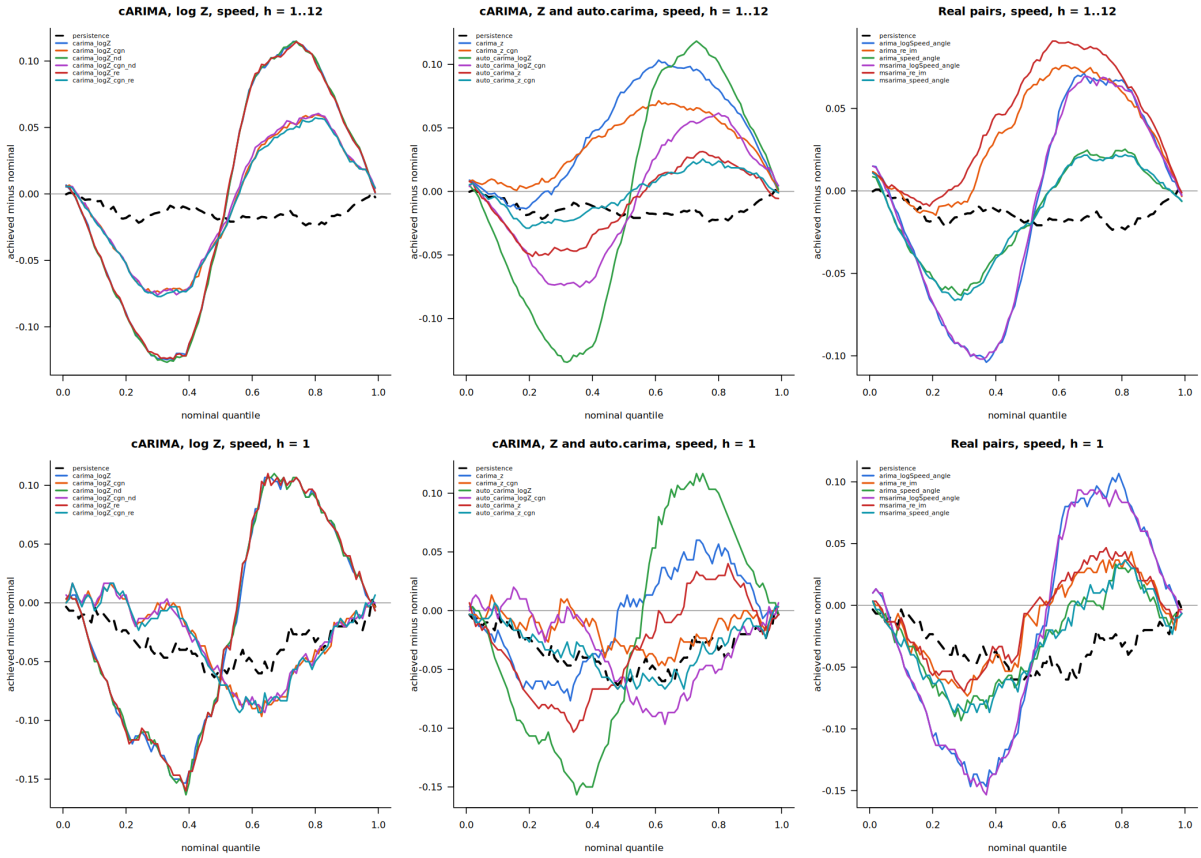


Figure 6: Calibration of speed: share of actual values below the forecast τ-quantile minus τ, against τ. Top: average over $h = 1$–12; bottom: $h = 1$. Persistence is the dashed black line in every panel.

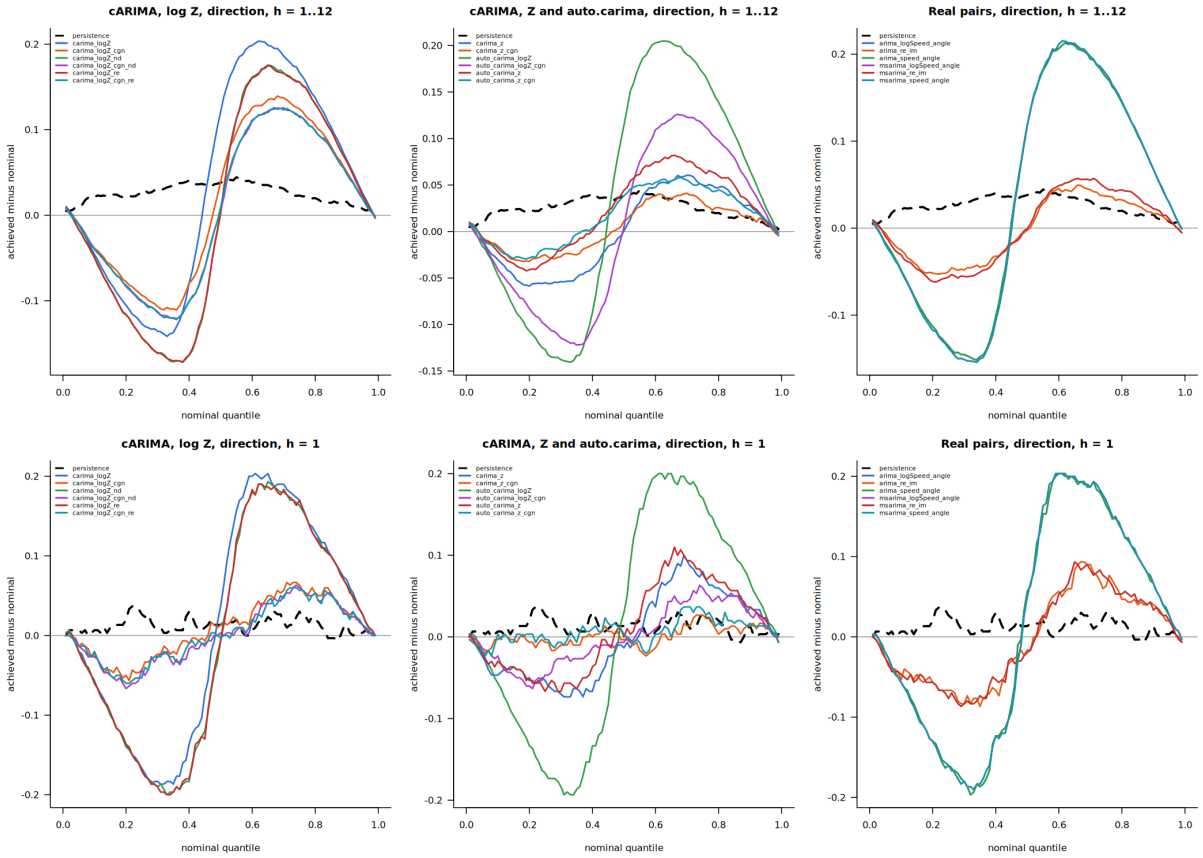


Figure 7: Calibration of direction, as in Figure 6.

8 Findings

1. **The cAR(0) orders were not a real feature of the data.** Cross-validation chooses $p > 0$ for 7 of 8 variants, but all orders forecast almost equally well (within 1%), on the cross-validation and on the test origins. The AR and MA terms nearly cancel, so the orders are weakly identified, and whether the chosen model has $p = 0$ depends on the window and on the optimiser. The order does not matter for forecasting up to 2 hours.

2. **Point forecasts up to 2 hours**: no model is more than 1% better than persistence over all horizons. The best are `auto_carima_z_cgn` and `carima_z_cgn` for the vector, `carima_logZ_re` and `carima_logZ_nd` for direction, and `arima_speed_angle` for speed; the cARIMA models for $\log Z$ with the generalised normal error are best at 10 and 30 minutes for speed and the vector.

3. **The real constant** (comment 4) gives the best direction forecasts of all models (27.10°), slightly ahead of no constant (27.14°); for the vector and speed it is about equal to persistence.

4. **Predictive distributions of direction: cARIMA for Z with the generalised normal error is the best model** (pinball 7% better than persistence, and better calibrated). It is also better than the two independent real models for $\operatorname{Re} Z$ and $\operatorname{Im} Z$, which ignore the correlation between the two errors.

5. **Predictive distributions of speed**: persistence, the real model for speed and cARIMA for Z with the generalised normal error are equally good.

6. **The models for $\log Z$ give good point forecasts but poor distributions**: their

intervals are much too wide in the centre. Changes of direction are large at low speeds and small at high speeds: the median 10-minute change is 11.0° at speeds up to 3 m/s, 2.6° at 5–8 m/s and 1.5° above 12 m/s. A model for $\log Z$ has one variance for the direction at all speeds, so at the usual speeds its direction intervals are too wide. In a model for Z, the same error of the wind vector is a large change of direction at low speed and a small one at high speed, as in the data.

7. **The generalised normal error improves all distributions**, most at short horizons. For all models, the central intervals are still too wide and the extreme ones slightly too narrow, because the actual changes over 2 hours are more sharply peaked than the sums of independent innovations the models assume. Only the bootstrap of persistence, which uses the empirical h-step changes directly, is close to perfectly calibrated.

9 Limitations and next steps

- One turbine and one year; no tests of the differences in accuracy (e.g. Diebold–Mariano); parameter uncertainty is ignored in the intervals.
- The explanation in finding 6 is not tested: the calibration could be computed separately for low and high speeds at the origin.
- The changes of wind are probably heteroscedastic (calm and turbulent periods), which would also explain why the actual 2-hour changes are more peaked than the model distributions. A model for the scale, or empirical intervals from the in-sample h-step errors of cARIMA (the `"empirical"` interval of `smooth`, not yet implemented in `complex`), may give better calibrated cARIMA distributions.
- The CRPS and an energy score for the joint distribution of speed and direction would evaluate the bivariate forecast directly, which is where a complex model should have an advantage.
- Not yet included: daily cycles (e.g. Fourier terms), VAR and VARMA benchmarks, forecasts of turbine power, and other datasets.

10 Reproducing the results

```
Rscript R/02_prepare_data.R     # needs data/raw/T1.csv
Rscript R/08_phase_plane.R
Rscript R/09_exp2_origins.R     # test and cross-validation origins
results/exp2/run_orders.sh      # R/10_all_orders.R (cv, test), R/11
results/exp2/run_forecasts.sh   # installs complex, R/12, R/13
```

Shared code is in `R/functions.R` and `R/exp2_functions.R`. The orders of the `auto_carima` models and of the real benchmarks are read from `results/tables/orders.csv` (first experiment). The `complex` package is the development version with the prediction intervals (branch `intervals`). On 32 cores with 30 workers, the cross-validation took 5.1 hours, the diagnostic on the test origins 3.0 hours, and the final forecasts of the 19 models 1.6 hours.

References

Novey, M., Adali, T. and Roy, A. (2010). A complex generalized Gaussian distribution: characterization, generation, and estimation. *IEEE Transactions on Signal Processing*, 58(3), 1427–1433.

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